

A Study on Semantic Information Retrieval

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Abstract

This paper presents a novel approach to semantic information retrieval in large-scale document databases. We utilize advanced text representation models and a hybrid search architecture to retrieve documents based on semantic content rather than simple keyword matches. Our experiments demonstrate significant improvements in retrieval performance over keyword-based baselines.

1. Introduction

Information retrieval systems have evolved from exact keyword matching to semantic search. Modern systems rely on deep language models that encode semantic meanings as vectors in high-dimensional spaces. Despite these advancements, extracting structured sections from scientific papers remains a challenging task due to formatting variations.

2. Proposed Methodology

Our model consists of three main components: a high-fidelity document parser, a semantic embedding encoder, and a hybrid vector-keyword retrieval engine. We describe the details of the document parser in this section, showcasing how structural cues like font sizes and text blocks are leveraged to map sections dynamically.

3. Experiments and Results

We evaluated our parsing pipeline on a benchmark of 100 computer science research articles. The results show that our font-size-based parser achieves a 95% classification accuracy across major sections, including Abstract, Introduction, and Conclusion. In contrast, standard rule-based tools achieve only 82%.

4. Conclusion

In this work, we developed a fast and robust PDF parsing engine. The system leverages font sizes and font weight metadata to divide documents into logical sections, paving the way for high-quality RAG pipelines. Future work will investigate table parsing and mathematical equation extraction.

References

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- [2] Smith, J. and Williams, A. 'Retrieval-Augmented Generation for Scientific Question Answering.' arXiv preprint, 2025.